

Project Proposal: Behavioral Dynamics and Alternative Data for Thin-File Credit Default Prediction

By Advik Goel (advik_goel@berkeley.edu), Faisal (faisal_k@berkeley.edu), Roman Hall (romanhall@berkeley.edu), Tung La (tung_la@berkeley.edu)

GitHub Repository: [DATASCI-207-FinalProject-Team1](#)

Motivation

Traditional credit scoring models rely on historical credit bureau records, including credit line length, revolving utilization, and delinquency marks. This reliance systematically disadvantages thin-file and underbanked populations, such as students and recent immigrants who lack credit records despite disciplined repayment habits. In modern lending, alternative behavioral proxies, including application trajectories and installment repayment dynamics, offer promising indicators of creditworthiness.

Our central question is whether alternative multi-table behavioral features can reliably predict loan default without bureau history. We hypothesize that (1) a hybrid cohort combining traditional and alternative signals outperforms traditional bureau features alone, and (2) this predictive gain is largest for thin-file applicants lacking bureau records. Validating these hypotheses demonstrates how fintechs can expand credit access while maintaining prudent portfolio risk controls.

Data

Our primary source is the Home Credit Default Risk dataset on Kaggle (<https://www.kaggle.com/competitions/home-credit-default-risk/data>): 307,511 loan applications for underbanked clients with an 8.07% default rate. The relational architecture joins five key tables on shared client and loan IDs:

1. application_train.csv: Demographics, income, assets, and external credit scores (EXT_SOURCE_1, 2, 3).
2. bureau.csv: External credit bureau debt and active credit lines (1.7M records).
3. previous_application.csv: Previous loan applications, approvals/refusals, and down payment terms (1.6M records).
4. installments_payments.csv: Repayment histories capturing installment amounts and payment delay gaps (13.6M records).
5. POS_CASH_balance.csv: Monthly cash loan balances capturing days past due (10.0M records).

We pre-aggregate relational records to one row per applicant for efficient execution.

Methodology

We frame default prediction as binary classification across three feature cohorts: traditional (demographic and bureau records), alternative (behavioral trajectories from previous applications, installments, and POS cash balances), and hybrid (combined feature space). We benchmark each cohort across three model families: baseline linear scorecards (Logistic

Regression and Decision Trees in scikit-learn), Gradient Boosted Decision Trees (LightGBM), and Deep Tabular Neural Networks in TensorFlow. Given class imbalance, we evaluate using ROC-AUC, PR-AUC, and F1 scores at calibrated thresholds. We run 5-fold stratified cross-validation and report metrics separately for applicants with and without bureau records to evaluate our second hypothesis.

Related Work

Recent studies show alternative data complements legacy bureau records. Berg et al. (2020) demonstrated that digital footprints predict consumer default with accuracy comparable to traditional credit bureau scores, particularly for thin-file borrowers. Similarly, Óskarsdóttir et al. (2019) demonstrated that integrating behavioral transaction trends into scoring algorithms boosts default discrimination for unbanked consumers. Our work expands these foundations by isolating multi-table behavioral trajectories across linear, boosted tree, and deep neural architectures.